

HARISH POTHAGONI

Data Engineer

Phone: +1 (302) 514-3132 | harishpothagoni6@gmail.com | linkedin.com/in/harish-p-0b1272289

PROFESSIONAL SUMMARY

Results-driven **Data Engineer** with **3 years of progressive project-based experience** supporting enterprise data platforms across **Banking and Healthcare domains**. Demonstrates strong hands-on expertise in **Python, SQL, ETL Pipelines, Data Warehousing, AWS, Apache Spark, Airflow, Snowflake, and Data Modeling**. Proven ability to architect scalable data pipelines, enforce data quality standards, optimize query and job performance, and partner effectively with analytics and business stakeholders in fast-paced Agile environments.

TECHNICAL SKILLS

Programming	Python, SQL
Data Engineering	ETL / ELT Pipelines, Data Ingestion, Data Transformation, Data Validation
Big Data	Apache Spark (PySpark), Hadoop (Basics)
Databases	Snowflake, PostgreSQL, MySQL, SQL Server
Cloud Platforms	AWS (S3, EC2, Lambda, Glue, Redshift)
Orchestration	Apache Airflow
Data Modeling	Star Schema, Snowflake Schema, Dimensional Modeling
Data Quality	Data Profiling, Reconciliation, Validation Checks
Version Control	Git, GitHub
Tools & IDEs	Jupyter Notebook, VS Code
Methodologies	Agile / Scrum

PROFESSIONAL EXPERIENCE

Data Engineer | Fifth Third Bank | Kentwood, MI

May 2024 – Present

Project: Enterprise Banking Data Lake & Analytics Platform

- Designed and developed scalable **ETL pipelines** using **Python and SQL** to ingest transactional and reference data from multiple upstream systems into centralized data stores.
- Built and optimized **PySpark-based data transformations** to process and aggregate large-scale banking datasets, significantly improving throughput and job reliability.
- Implemented robust data ingestion workflows from diverse sources including relational databases (PostgreSQL, SQL Server) and structured flat files into the enterprise data lake.
- Developed and maintained an **AWS S3-based data lake architecture** capable of handling structured and semi-structured data at scale with partitioned storage strategies.
- Automated end-to-end workflows using **Apache Airflow DAGs** to ensure reliable scheduling, dependency management, and real-time monitoring of production pipelines.
- Designed **Snowflake data warehouse models** using star and snowflake schemas aligned with business KPIs, enabling efficient reporting and self-service analytics.
- Implemented comprehensive **data quality checks and reconciliation frameworks**, reducing data discrepancies and improving downstream analytical accuracy.
- Optimized complex SQL queries and PySpark jobs through partitioning, caching, and indexing strategies — achieving measurable reductions in pipeline latency.
- Collaborated closely with BI and analytics teams to define data contracts and deliver clean, well-documented, analytics-ready datasets.
- Enforced data security and compliance standards using IAM-based access controls, S3 bucket policies, and at-rest/in-transit encryption in regulated banking environments.
- Authored and maintained technical documentation covering pipeline design, schema definitions, lineage, and operational runbooks.

Environment: *Python, SQL, PySpark, AWS (S3, EC2, Glue), Apache Airflow, Snowflake, Git, Linux*

Junior Data Engineer | Sonata Software | Hyderabad, India

June 2021 – August 2023

Project: Healthcare Data Integration & Reporting Platform

- Assisted in designing and building **data ingestion and ETL workflows** using **Python and SQL** to consolidate structured healthcare datasets from heterogeneous source systems.
- Integrated data from multiple channels — including RDBMS databases, CSV/Excel flat files, and REST APIs into centralized staging and reporting repositories.
- Performed systematic **data cleansing, transformation, and validation** using Python scripts and SQL stored procedures, significantly improving source data reliability.
- Supported development and maintenance of **data warehouse tables and dimensional models** enabling consistent reporting and analytics use cases across business units.
- Created optimized SQL queries, views, and materialized tables to accelerate downstream analytics and reduce report generation time.
- Implemented **PySpark transformation jobs** for batch processing of high-volume healthcare records, ensuring timely delivery of analytics-ready data.
- Ensured strict compliance with **HIPAA healthcare data standards** through secure data handling, masking of PII fields, and audit-ready validation workflows.
- Actively monitored scheduled pipeline jobs, responded to execution failures, and performed root-cause analysis to minimize data delivery delays.
- Supported senior engineers in diagnosing and resolving complex data quality issues, pipeline bottlenecks, and schema drift across integration layers.
- Produced detailed ETL workflow documentation, data mapping specs, and source-to-target lineage diagrams for client review and internal knowledge sharing.
- Built foundational expertise in enterprise data engineering patterns including medallion architecture, SCD handling, and incremental load strategies.

Environment: *Python, SQL, PySpark, AWS (S3), Data Warehousing, Git, Linux*

EDUCATION

Wilmington University

Master of Science — Information Systems and Technology
New Castle, DE | GPA: 3.8 / 4.0

Jan 2024 – May 2025

Andhra University

Bachelor of Science — Computer Science
Visakhapatnam, India

June 2017 – May 2021

CERTIFICATIONS

- AWS Certified Data Analytics – Specialty (Planned)
- Apache Spark & PySpark – Udemy
- SQL for Data Engineering – Coursera
- Python for Data Engineering – Udemy

Harish Pothagoni • Data Engineer • harishpothagoni6@gmail.com • +1 (302) 514-3132